

BASIC ANTIVIRUS SIMULATION

Internship Project Report

Submitted By

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Course: B.Tech Computer Science Engineering (CSE)

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Certificate

This is to certify that the project titled "**Basic Antivirus Simulation**" has been successfully completed as part of the Cyber Security Internship program.

Acknowledgement

I would like to express my sincere gratitude to my internship mentor and organization for providing me with the opportunity to work on this project. This project helped me understand the basic concepts of malware detection, file scanning, and quarantine mechanisms used in antivirus software.

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1. Introduction

Antivirus software is an important component of cybersecurity that helps detect and remove malicious files from computer systems. This project demonstrates a basic antivirus simulation using Python. The program scans files, identifies malicious signatures, quarantines infected files, and generates a scan report.

2. Objective

The main objectives of this project are:

- To understand the working of antivirus software.
- To perform signature-based malware detection.
- To scan files for predefined malicious signatures.
- To quarantine infected files.
- To generate a detailed scan report.

3. Tools and Technologies Used

Tool	Purpose
Python	Programming Language
VS Code	Code Editor
File Handling	Reading and Writing Files
OS Module	Directory Operations
Shutil Module	Moving Files to Quarantine

4. Methodology

The project works through the following steps:

Step 1: Load Virus Signatures

The antivirus loads predefined malware signatures from the file:

virus_signatures.txt

Step 2: Scan Files

The program scans all files present inside the TestFiles directory.

Step 3: Detect Threats

If any malware signature is found inside a file, it is marked as infected.

Step 4: Quarantine File

The infected file is moved to the Quarantine folder.

Step 5: Generate Report

A scan report is automatically generated and saved as:

scan_report.txt

5. Project Structure

Basic_Antivirus_Simulation

```
├── antivirus.py
├── virus_signatures.txt
├── scan_report.txt
├── README.md
├── TestFiles
└── Quarantine
```

6. Implementation

Virus Signatures Used

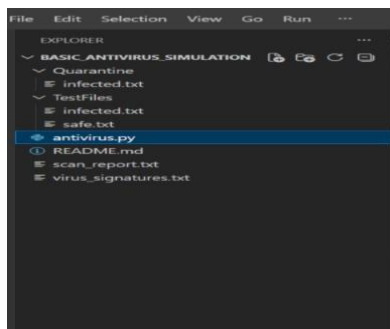
- trojan
- virus
- malware
- worm
- keylogger
- ransomware

Features Implemented

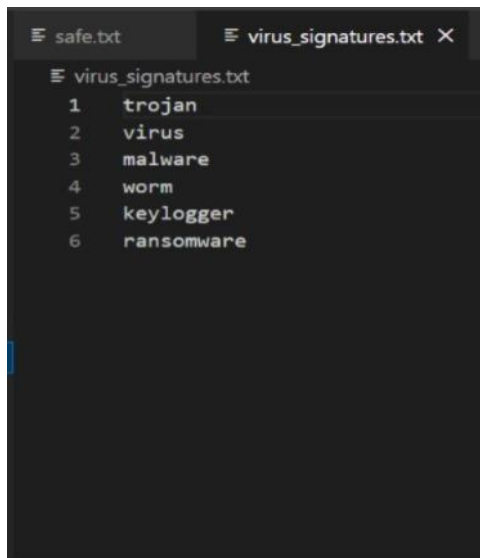
- File Scanning
 - Malware Detection
 - Quarantine System
 - Report Generation
-

7. Results and Output

Screenshot 1: Project Folder Structure

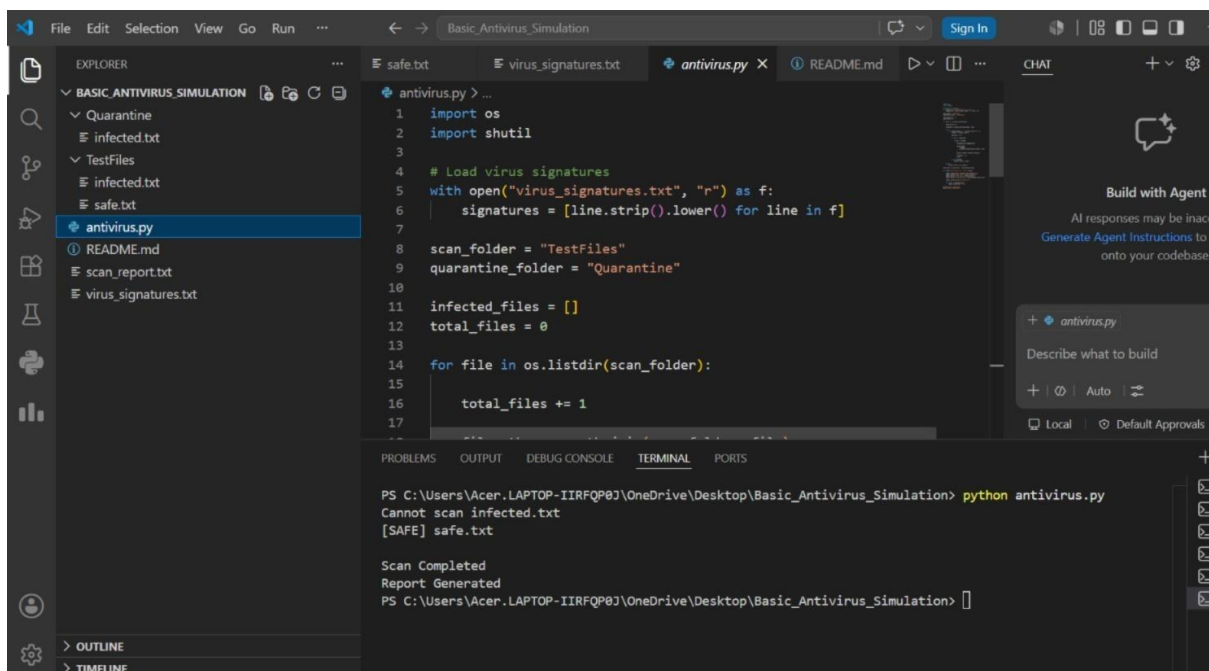


Screenshot 2: Virus Signature File

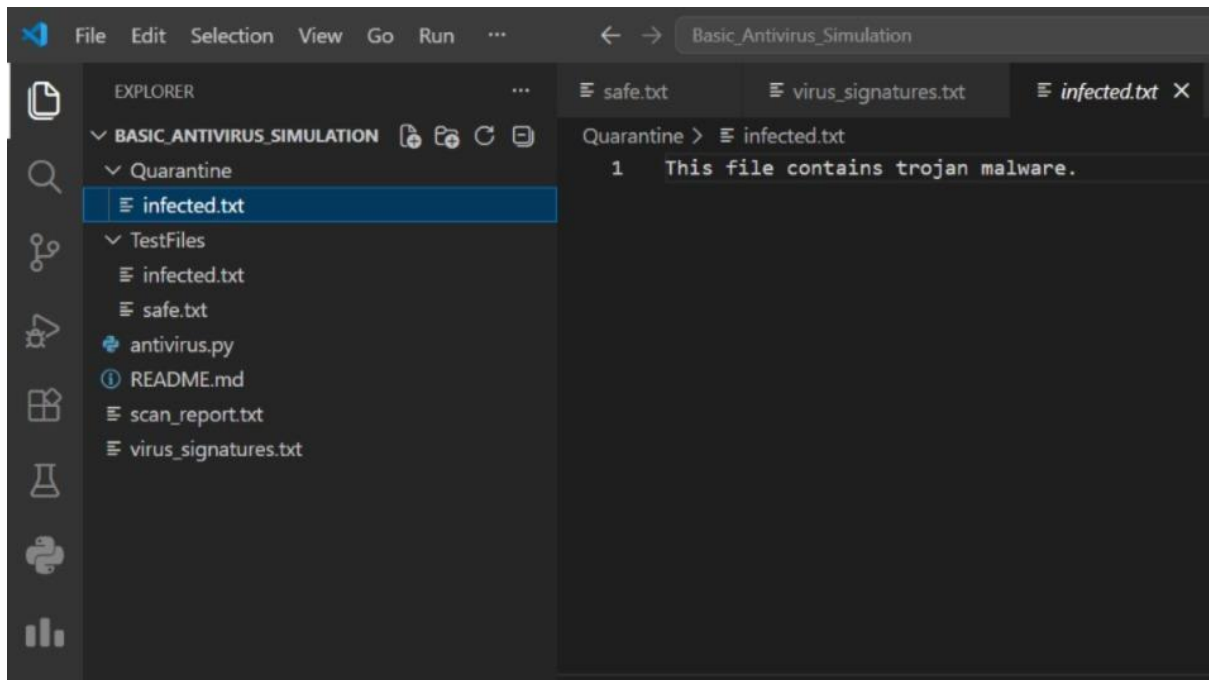


```
safe.txt virus_signatures.txt X
virus_signatures.txt
1 trojan
2 virus
3 malware
4 worm
5 keylogger
6 ransomware
```

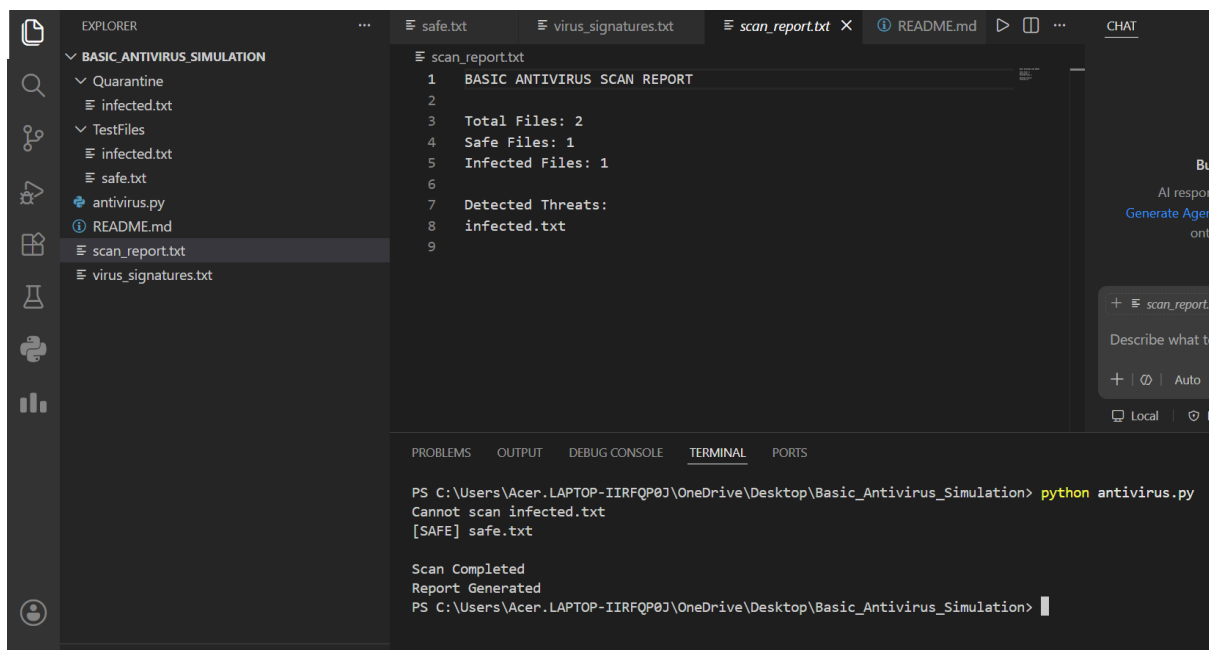
Screenshot 3: Program Execution in VS Code



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File Edit Selection View Go Run ... Basic_Antivirus_Simulation Sign In
EXPLORER
BASIC_ANTIVIRUS_SIMULATION
  Quarantine
  infected.txt
  TestFiles
    infected.txt
    safe.txt
  antivirus.py
  README.md
  scan_report.txt
  virus_signatures.txt
antivirus.py X README.md
1 import os
2 import shutil
3
4 # Load virus signatures
5 with open("virus_signatures.txt", "r") as f:
6     signatures = [line.strip().lower() for line in f]
7
8 scan_folder = "TestFiles"
9 quarantine_folder = "Quarantine"
10
11 infected_files = []
12 total_files = 0
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14 for file in os.listdir(scan_folder):
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16     total_files += 1
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Screenshot 5: Generated Scan Report



Sample Output

[SAFE] safe.txt

[THREAT DETECTED] infected.txt

Scan Completed

Report Generated

8. Conclusion

The Basic Antivirus Simulation project was successfully developed using Python. The application demonstrated how antivirus software performs signature-based malware detection, quarantines infected files, and generates reports. This project helped in understanding fundamental cybersecurity concepts and antivirus mechanisms.

9. Future Scope

The project can be enhanced by adding:

- Graphical User Interface (GUI)
 - Real-time File Monitoring
 - Hash-Based Malware Detection
 - Cloud-Based Threat Database
 - Email Alerts for Threat Detection
 - Multi-threaded Scanning
-

References

1. Python Official Documentation
 2. Cyber Security Fundamentals
 3. Malware Analysis Concepts
 4. File Handling in Python
-

Declaration

I hereby declare that this project report titled "**Basic Antivirus Simulation**" is my original work and has been completed as part of my Cyber Security Internship.

Name: Atharv Ansh Singh

Date: 22/06/2026